GITAM (Deemed to be University)
Bengaluru Campus
Department of Computer Science and Systems Engineering

DISCOVERING MENTAL DISORDERS LEVERAGING CONTRASTIVE LEARNING

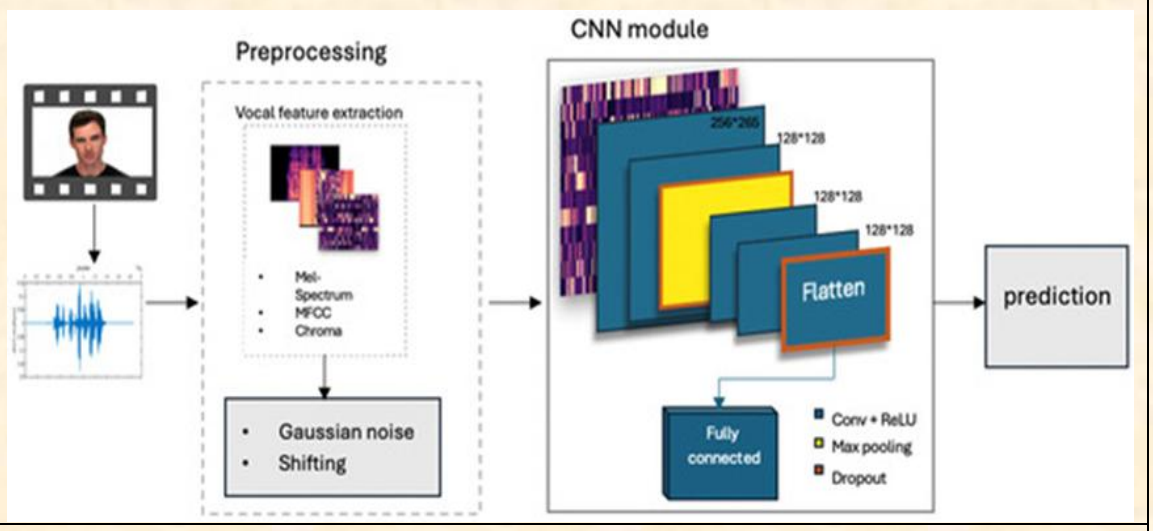
Batch No. CSE-91

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Abstract:

This project presents an AI-based system for detecting mental health conditions using speech emotion recognition. The system analyses speech signals and extracts important audio features such as MFCC and Mel-Spectrogram. These features are used as input to a Convolutional Neural Network (CNN) for emotion classification. Contrastive learning is applied to improve the accuracy and performance of the model. The system generates emotion probability results and an automated report for mental health analysis. This approach helps in early detection and monitoring of mental health conditions.

Design Strategies

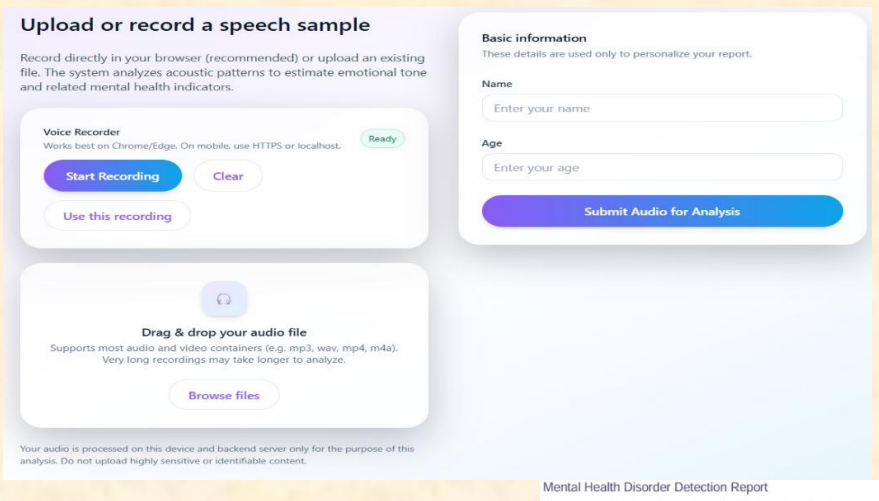


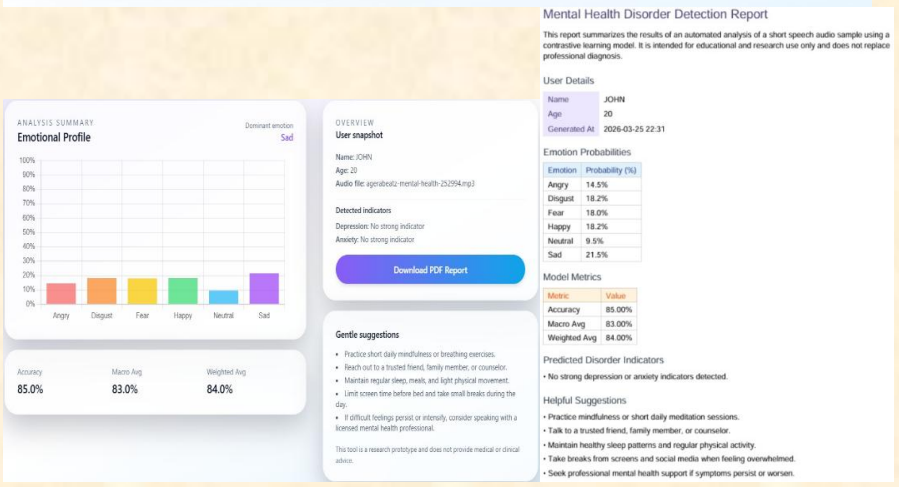
Objective:

- Design and develop a speech-based mental health detection system
- Perform speech signal preprocessing and noise reduction
- Extract discriminative audio features using MFCC and Mel-Spectrogram
- Train a Convolutional Neural Network (CNN) for speech emotion classification
- Implement contrastive learning for feature embedding and representation learning
- Use contrastive loss to improve class separability and reduce intra-class variance
- Perform emotion probability estimation for mental health assessment
- Develop an automated mental health analysis and reporting system
- Evaluate model performance using classification accuracy and loss metrics
- Enable early detection and continuous monitoring of mental health conditions

Results:







Identification of Tools / Algorithms / Technologies

Python was used for implementation. Librosa was used for audio processing and feature extraction. MFCC and Mel-Spectrogram were used as input features. CNN was used as the classification algorithm, and contrastive learning was used to improve model accuracy.

Conclusion:

This project presents a speech-based mental health detection system using feature extraction and deep learning techniques. The CNN model classifies emotional patterns, and contrastive learning improves the model performance. The system provides an effective method for mental health analysis and early detection.

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